

Luca Cazzola

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Summary

Applied Scientist focused on *Computer Vision* and *Generative Models* for robotic and digital motion. Author of multiple publications across international conferences, spanning diffusion-based motion synthesis and human activity recognition. I now bridge research-level knowledge into industrial applications, from perception systems for manufacturing partners to robot learning pipelines.

Publications

BlendAnything: A Blender Plugin for Cross-Topology Motion Blending [\(Website\)](#) | *SIGGRAPH Posters 2026*

- A plugin enabling motion generative models within Blender: the open-source 3D editor.
- Accepted as poster in the best conference world-wide for computer graphics, fitting the top 20% of applicants.

Neural Motion Blending Across Arbitrary Character Topologies [\(Website\)](#) | *CGI 2026*

- Built diffusion based generative pipeline to repurpose motion between incompatible skeleton hierarchies.
- This work is the first of its kind and opens an entirely new branch of research.

Kinetic Mining in Context: Few-Shot Action Synthesis via Text-to-Motion Distillation [\(Website\)](#) | *ICPR 2026*

- Implemented and evaluated *multimodal diffusion generative models* for human motion synthesis and activity recognition.
- Achieved a *+23% accuracy gain* in data-scarce human activity recognition through LoRA-based parameter-efficient fine tuning.

Experience

Fondazione Bruno Kessler – Applied Scientist | Trento, Italy *Jun 2026 – Ongoing*

- Building teleoperation systems for Vision-Language-Action data collection, and experimenting with recent VLA architectures for robotic manipulation.
- Designing end-to-end computer vision pipelines for large-scale manufacturing partners, integrating PLC data, a CVAT-based annotation workflow for operators, and a DINO self-supervised pretraining and fine-tuning strategy to automate artifact detection on the production line and cut costs.

Awards

Industrial AI Challenge (1st Place) | Hub Innovazione Trentino *Sep 2024 – Dec 2024*

- Led a 7-person team to 1st place in an *AWS* sponsored competition, managing industrial partner relations and internal operations while contributing to the software development for solving a multi-machine industrial scheduling problem.
- Engineered an AI enhanced industrial scheduling pipeline featuring integer constraint programming and genetic algorithms, providing the foundational methodology for a GECCO 2025 publication, validating its novelty at a top-tier conference.

Projects

Robotic Manipulator: Perception, Control, and ROS Integration [\(Project page\)](#) | 2023

- Developed an end-to-end perception-to-control pipeline for a *UR5 robotic arm*, bridging vision and motion control.
- Designed a robust Sim-to-Real workflow by training a *YOLO* object detector on a custom *synthetic dataset* in Blender.
- Optimized a custom 6D object pose estimation system using point cloud interpolation.

GPU Parallelism: CUDA, Matrix Transposition and 2D Convolution [\(Project page\)](#) | 2024

- Optimized C++ CUDA kernels for 2D convolutions, achieving 341 GB/s effective bandwidth (76% of theoretical peak) by implementing shared memory tiling and memory coalescing.

Skills

AI & CS: Machine Learning, Deep Learning, Computer Vision, Diffusion Models, LLM, VLM, Motion Synthesis, Gen AI, Algorithms

Technical: Python, C, C++, PyTorch, OpenCV, ROS, CUDA, Docker, Google OR-Tools, Git, Blender, LaTeX

Languages: Italian (Native), English (Fluent)

Education

Master's Degree – Artificial Intelligence Systems – University of Trento

2023 – 2026

Bachelor's Degree – Computer Science – University of Trento

2020 – 2023